

Mathematical Statistics Final Cheat Sheet

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1. Mathematical Tools

Differentiation Rules:

- Product: $(uv)' = u'v + uv'$
- Quotient: $(\frac{u}{v})' = \frac{u'v - uv'}{v^2}$
- Chain: $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$
- Logarithm: $\frac{d}{dx} \ln(f(x)) = \frac{f'(x)}{f(x)}$
- Exponential: $\frac{d}{dx} e^{f(x)} = f'(x)e^{f(x)}$

Integration by Parts:

$$\int u dv = uv - \int v du$$

- Choice order (L.I.A.T.E): Log, Inverse trig, Algebraic, Trig, Exponential

Gamma Function:

- $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
- Property: $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$
- Integer: $\Gamma(n) = (n-1)!$

1.5 Key Transformations

Transformation Formula (Univariate)

If $Y = g(X)$ is monotonic:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

For many-to-one (e.g., $Y = X^2$), sum over roots

Jacobian (Bivariate)

For $(U, V) = g(X, Y)$:

$$f_{U,V}(u, v) = f_{X,Y}(x, y) |J|$$

where $J = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$

Order Statistics

For i.i.d. $X_1, \dots, X_n$ with CDF F , PDF f :

Minimum: $f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1}f(x)$

Maximum: $f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$

k -th: $f_{X_{(k)}}(x) =$

$$\frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

Sums of Independent RVs:

- $X_i \sim \text{Exp}(\lambda)$ iid: $\sum X_i \sim \text{Gamma}(n, \lambda)$
- $X_i \sim \text{Poisson}(\lambda_i)$: $\sum X_i \sim \text{Poisson}(\sum \lambda_i)$
- $X_i \sim N(\mu_i, \sigma_i^2)$:
 $\sum a_i X_i \sim N(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2)$
- $X_i \sim \text{Binomial}(n_i, p)$:
 $\sum X_i \sim \text{Binomial}(\sum n_i, p)$

2. Sampling Distributions

Law of Large Numbers (LLN)

If $X_1, X_2, \dots$ i.i.d. with $\mathbb{E}[X_i] = \mu$,

$\text{Var}(X_i) = \sigma^2 < \infty$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Chebyshev proof: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2/n}{\varepsilon^2} \rightarrow 0$

Central Limit Theorem (CLT)

If X_i i.i.d. with mean μ and variance $\sigma^2 > 0$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Equivalently: $\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n

For sums: $S_n = \sum X_i \approx N(n\mu, n\sigma^2)$

Continuity Correction (Discrete $\rightarrow$ Continuous)

For $X \sim B(n, p)$:

$$P(X \leq k) \approx P\left(Z \leq \frac{k+0.5-np}{\sqrt{np(1-p)}}\right)$$

Rule: Normal approx reliable if $np(1-p) \gtrsim 10$

Chi-Square Distribution: $\chi^2(k)$

If $Z_1, \dots, Z_k \sim N(0, 1)$ independent:

$$\sum_{i=1}^k Z_i^2 \sim \chi^2(k)$$

PDF: $f(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{k/2-1} e^{-x/2}, x > 0$

$\mathbb{E}[\chi^2(k)] = k, \text{Var}(\chi^2(k)) = 2k$

MGF: $M(t) = (1 - 2t)^{-k/2}, t < 1/2$

Additivity: If $X \sim \chi^2(k_1), Y \sim \chi^2(k_2)$ independent:

$$X + Y \sim \chi^2(k_1 + k_2)$$

t-Distribution: $t(n)$

If $Z \sim N(0, 1), V \sim \chi^2(n)$ independent:

$$T = \frac{Z}{\sqrt{V/n}} \sim t(n)$$

$\mathbb{E}[T] = 0$ ($n > 1$), $\text{Var}(T) = \frac{n}{n-2}$ ($n > 2$)

As $n \rightarrow \infty$: $t(n) \rightarrow N(0, 1)$

F-Distribution: $F(n_1, n_2)$

If $V_1 \sim \chi^2(n_1), V_2 \sim \chi^2(n_2)$ independent:

$$F = \frac{V_1/n_1}{V_2/n_2} \sim F(n_1, n_2)$$

$\mathbb{E}[F] = \frac{n_2}{n_2-2}$ ($n_2 > 2$)

If $F \sim F(n_1, n_2)$: $\frac{1}{F} \sim F(n_2, n_1)$

If $T \sim t(n)$: $T^2 \sim F(1, n)$

Normal Sample: Key Results

If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ i.i.d.:

- $\bar{X} \sim N(\mu, \sigma^2/n)$
- $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$ where
 $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- $\bar{X}$ and S^2 are independent
- $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ (σ known)
- $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$ (σ unknown)

Delta Method

If $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, \tau^2)$ and $g'(\theta) \neq 0$:

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \Rightarrow N(0, [g'(\theta)]^2 \tau^2)$$

Variance: $\text{Var}(g(\hat{\theta}_n)) \approx [g'(\theta)]^2 \text{Var}(\hat{\theta}_n)$

Key Distribution Relationships:

- $Z^2 \sim \chi^2(1)$ if $Z \sim N(0, 1)$
- $\chi^2(k_1) + \chi^2(k_2) = \chi^2(k_1 + k_2)$ (indep)
- $T^2 \sim F(1, n)$ if $T \sim t(n)$
- $1/F \sim F(n_2, n_1)$ if $F \sim F(n_1, n_2)$
- As $n \rightarrow \infty$: $t(n) \rightarrow N(0, 1)$

3. Point Estimation

Estimator Properties

Bias: $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$

Variance: $\text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$

MSE: $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$

Key Result: Unbiased $\Rightarrow$ MSE = Var

Consistency

$\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$ (converges in probability)

Sufficient conditions:

- Unbiased + $\text{Var}(\hat{\theta}_n) \rightarrow 0$
- $\text{MSE}(\hat{\theta}_n) \rightarrow 0$
- Bias($\hat{\theta}_n$) $\rightarrow 0$ and $\text{Var}(\hat{\theta}_n) \rightarrow 0$

Relative Efficiency

For unbiased $\hat{\theta}_1, \hat{\theta}_2$:

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

$\text{eff} > 1$: $\hat{\theta}_1$ more efficient

Common Estimator Biases

- Sample variance (MLE):
 $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ is **biased**
 - $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \frac{n-1}{n} \sigma^2$ (underestimates)
 - Unbiased version:
 $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- Uniform(0, θ): $\hat{\theta} = X_{(n)}$ is biased
 - $\mathbb{E}[X_{(n)}] = \frac{n}{n+1} \theta$ (underestimates)
 - $\mathbb{E}[X_{(1)}] = \frac{\theta}{n+1}$ (min)
 - Unbiased: $\frac{n+1}{n} X_{(n)}$

Variance Formulas:

- Linear combo: $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$
- Independent:
 $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- General: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Method of Moments (MoM)

Match sample moments to population moments:

- Set $\bar{X} = \mathbb{E}[X]$ (first moment)
- Set $\frac{1}{n} \sum X_i^2 = \mathbb{E}[X^2]$ (second moment)
- Solve for θ

Example: Gamma(α, β):

- $\mathbb{E}[X] = \alpha\beta = \bar{X}$
- $\text{Var}(X) = \alpha\beta^2 = S^2$
- Solve: $\hat{\beta} = S^2/\bar{X}, \hat{\alpha} = \bar{X}^2/S^2$

Common Pitfalls:

- S^2 (unbiased) has $n-1$; MLE $\hat{\sigma}^2$ has n
- t-distribution df is $n-1$ (not n !)
- For χ^2 variance test: $(n-1)S^2/\sigma^2$
- Type I error = α , Type II error = β

Maximum Likelihood (MLE)

Likelihood: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$

Log-likelihood: $\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$

MLE: $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$

Finding MLE: Solve $\frac{d\ell}{d\theta} = 0$ (score equation)

Common MLEs:

- Bernoulli(p): $\hat{p} = \bar{X}$
- Poisson(λ): $\hat{\lambda} = \bar{X}$
- Normal(μ, σ^2): $\hat{\mu} = \bar{X}, \hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$
- Exponential(λ): $\hat{\lambda} = \frac{1}{\bar{X}}$
- Uniform(0, θ):
 $\hat{\theta} = X_{(n)} = \max\{X_1, \dots, X_n\}$

MLE Finding Tips:

- Support depends on θ :* Uniform(0, θ) or (a, b)
 - MLE is at boundary: $\max\{X_i\}$ or $\{\min\{X_i\}, \max\{X_i\}\}$
 - Don't differentiate!**
- Log tricks:* $\frac{d}{d\theta} \log f = \frac{f'}{f}, \log(ab) = b \log a$

- Check second derivative to verify maximum

MLE Properties:

- Invariance: If $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$
- Consistent under regularity conditions
- Asymptotically efficient
- Asymptotic normality: $\hat{\theta}_n \xrightarrow{d} N(\theta, \frac{1}{nI(\theta)})$

Fisher Information $I(\theta)$

Definition: Measures information about θ from one observation.

Two equivalent forms:

1. Expected curvature:
$$I(\theta) = -\mathbb{E} \left[\frac{d^2 \log f(X; \theta)}{d\theta^2} \right]$$
2. Variance of score:
$$I(\theta) = \mathbb{E} \left[\left(\frac{d \log f(X; \theta)}{d\theta} \right)^2 \right]$$

Relationship: Both equal under regularity conditions.

How to compute $I(\theta)$:

1. Write $\log f(x; \theta)$
2. Find score: $S(\theta) = \frac{d \log f}{d\theta}$
3. Method 1: $I(\theta) = \mathbb{E}[S(\theta)^2]$
4. Method 2: Find $\frac{d^2 \log f}{d\theta^2}$, then
$$I(\theta) = -\mathbb{E} \left[\frac{d^2 \log f}{d\theta^2} \right]$$

Sample Fisher Information:

For n i.i.d. observations: $nI(\theta)$

Total log-likelihood: $\ell(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$

Sample info: $I_n(\theta) = -\mathbb{E} \left[\frac{d^2 \ell(\theta)}{d\theta^2} \right] = nI(\theta)$

MLE Variance (Cramer-Rao):

$$\text{Var}(\hat{\theta}_{\text{MLE}}) \approx \frac{1}{nI(\hat{\theta})}$$

Estimated variance: $\widehat{\text{Var}}(\hat{\theta}) = \frac{1}{nI(\hat{\theta})}$

Common Fisher Information:

- Bernoulli(p): $I(p) = \frac{1}{p(1-p)}$
- Poisson(λ): $I(\lambda) = \frac{1}{\lambda}$
- $N(\mu, \sigma^2)$ (known σ^2): $I(\mu) = \frac{1}{\sigma^2}$
- $N(\mu, \sigma^2)$ (known μ): $I(\sigma^2) = \frac{1}{2\sigma^4}$
- Exponential(λ): $I(\lambda) = \frac{1}{\lambda^2}$

Example: Bernoulli(p) - Full derivation

Step 1: Single observation Fisher info

$$\begin{aligned} f(x; p) &= p^x(1-p)^{1-x} \quad (\text{one obs}) \\ \log f &= x \log p + (1-x) \log(1-p) \\ \frac{d \log f}{dp} &= \frac{x}{p} - \frac{1-x}{1-p} \\ \frac{d^2 \log f}{dp^2} &= -\frac{x}{p^2} - \frac{1-x}{(1-p)^2} \\ I(p) &= -\mathbb{E} \left[-\frac{X}{p^2} - \frac{1-X}{(1-p)^2} \right] \\ &= \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p(1-p)} \end{aligned}$$

Step 2: Sample of size n

For $X_1, \dots, X_n$ i.i.d.: $nI(p) = \frac{n}{p(1-p)}$

Step 3: MLE variance

$$\text{Var}(\hat{p}) = \frac{1}{nI(\hat{p})} = \frac{p(1-p)}{n}$$

Large Sample CI for MLE:

Generic: $\hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{1}{nI(\hat{\theta})}}$

For proportion: $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

4. Confidence Intervals

Definition

$(1-\alpha)100\%$ CI for θ : Random interval $[L, U]$ with

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Interpretation: $(1-\alpha)100\%$ of intervals contain true θ .

Pivotal Quantity

A function $Q(X_1, \dots, X_n; \theta)$ whose distribution does not depend on θ .

Examples: $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$, $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$

To construct CI: Find a, b with $P(a \leq Q \leq b) = 1 - \alpha$, then solve for θ .

Common CIs

Mean (σ known, large n):

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Mean (σ unknown, Normal):

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Proportion (large n):

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Variance (Normal):

$$\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}$$

Ratio of Variances:

$$\frac{s_1^2/s_2^2}{F_{\alpha/2, n_1-1, n_2-1}} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} \cdot F_{\alpha/2, n_1-1, n_2-1}$$

Two-Sample Mean (equal var):

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

Sample Size Determination

For mean with margin E and confidence $(1-\alpha)$:

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

For proportion:

$$n = \hat{p}(1-\hat{p}) \left(\frac{z_{\alpha/2}}{E} \right)^2$$

Conservative (no prior $\hat{p}$): Use $\hat{p} = 0.5$ (gives largest n)

5. Hypothesis Testing

Framework

Null hypothesis H_0 : Status quo (assume true)

Alternative H_1 : What we want evidence for

Types:

- Two-sided: $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$
- Upper: $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$
- Lower: $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$

Error Types

Type I Error (False Positive):

$\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$

Type II Error (False Negative):

$\beta = P(\text{Fail reject } H_0 | H_1 \text{ true})$

Power: $1 - \beta = P(\text{Reject } H_0 | H_1 \text{ true})$

p-Value

Probability under H_0 of observing test statistic as extreme or more extreme.

Decision: Reject H_0 if $p \leq \alpha$

Interpretation:

- $p \leq 0.01$: Very strong evidence against H_0
- $0.01 < p \leq 0.05$: Moderate evidence
- $0.05 < p \leq 0.10$: Weak evidence
- $p > 0.10$: Little/no evidence

Common Test Statistics

Z-test (mean, σ known):

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ under } H_0$$

t-test (mean, σ unknown):

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

Proportion test (large n):

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0, 1) \text{ under } H_0$$

Chi-square (variance):

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi_{n-1}^2 \text{ under } H_0$$

Two-sample t-test (equal var):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2} \text{ under } H_0$$

where $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

F-test (variance ratio):

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1} \text{ under } H_0 : \sigma_1^2 = \sigma_2^2$$

Rejection Regions

Two-sided:

- Z or t: $|T| > t_{\alpha/2}$
- χ^2 : $\chi^2 < \chi_{1-\alpha/2}^2$ or $\chi^2 > \chi_{\alpha/2}^2$
- F: $F < F_{1-\alpha/2}$ or $F > F_{\alpha/2}$

One-sided (upper):

- Z or t: $T > t_{\alpha}$
- χ^2 : $\chi^2 > \chi_{\alpha}^2$
- F: $F > F_{\alpha}$

Power & Sample Size

Power function: $\pi(\theta) = P(\text{Reject } H_0 | \theta)$

One-sided (upper): $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$

$$\pi(\mu_1) = \Phi \left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} - z_{\alpha} \right)$$

Two-sided: Replace z_{α} with $z_{\alpha/2}$

Sample size: For power $1 - \beta$ at μ_1 :

$$n = \left(\frac{(z_{\alpha} + z_{\beta})\sigma}{\mu_1 - \mu_0} \right)^2$$

Two-sided: $n = \left(\frac{(z_{\alpha/2} + z_{\beta})\sigma}{\mu_1 - \mu_0} \right)^2$

Test Decision Rules

p-value approach:

- Calculate test statistic
- Find p-value
= $P(\text{as extreme or more } | H_0)$
- Reject if $p \leq \alpha$

Critical value approach:

- Find critical value(s) from distribution
- Reject if test statistic in rejection region

p-Value Calculation

Upper-tail: $p = P(T \geq t_{\text{obs}} | H_0)$

Lower-tail: $p = P(T \leq t_{\text{obs}} | H_0)$

Two-tailed: $p = 2P(|T| \geq |t_{\text{obs}}| | H_0)$

CI-Testing Duality

$(1-\alpha)$ CI $[L, U]$ corresponds to level- α test:

- Reject $H_0 : \theta = \theta_0$ iff $\theta_0 \notin [L, U]$

- Fail to reject iff $\theta_0 \in [L, U]$

6. Likelihood Ratio Tests

Neyman-Pearson Lemma

For simple $H_0 : \theta = \theta_0$ vs simple $H_1 : \theta = \theta_1$:
Most powerful test of size α rejects when:

$$\Lambda(x) = \frac{L(\theta_0; x)}{L(\theta_1; x)} \leq k$$

where k satisfies $P(\Lambda \leq k | H_0) = \alpha$

General LRT

For composite hypotheses $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$:

Parameter Spaces:

- Ω = full parameter space (unrestricted)
- Ω_0 = restricted parameter space under H_0

Likelihood ratio:

$$\lambda(x) = \frac{\sup_{\theta \in \Omega_0} L(\theta; x)}{\sup_{\theta \in \Omega} L(\theta; x)} = \frac{L(\hat{\theta}_0; x)}{L(\hat{\theta}; x)}$$

where $\hat{\theta}_0$ = MLE under H_0 (restricted), $\hat{\theta}$ = MLE (unrestricted)

Properties: $0 \leq \lambda(x) \leq 1$, smaller $\lambda \rightarrow$ stronger evidence against H_0

Wilks' Theorem

Under H_0 and regularity conditions:

$$-2 \log \lambda(X) \xrightarrow{d} \chi_\nu^2$$

where $\nu = \dim(\Theta) - \dim(\Theta_0)$ (df = difference in parameters)

Rejection: $-2 \log \lambda(x) > \chi_{\alpha, \nu}^2$

LRT Calculation Steps

1. Find unrestricted MLE:
 $\hat{\theta} = \arg \max_{\theta \in \Omega} L(\theta)$
2. Find restricted MLE:
 $\hat{\theta}_0 = \arg \max_{\theta \in \Omega_0} L(\theta)$
3. Compute: $\lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta})}$
4. Test statistic: $-2 \log \lambda$
5. Compare to χ_ν^2 critical value

Common LRT Scenarios

Normal mean (σ^2 known): $H_0 : \mu = \mu_0$

- $\lambda = \exp \left\{ -\frac{n(\bar{x} - \mu_0)^2}{2\sigma^2} \right\}$
- $-2 \log \lambda = \frac{n(\bar{x} - \mu_0)^2}{\sigma^2} \sim \chi_1^2$
- Equivalent to Z-test

Normal variance (μ known): $H_0 : \sigma^2 = \sigma_0^2$

- $\lambda = \left(\frac{\hat{\sigma}^2}{\sigma_0^2} \right)^{n/2} e^{-\frac{n}{2} \left(\frac{\hat{\sigma}^2}{\sigma_0^2} - 1 \right)}$
- $-2 \log \lambda \sim \chi_1^2$

Two Normal means (equal var): $H_0 : \mu_1 = \mu_2$

- Simplifies to t-test
- $\nu = 1$ (one constraint)

LRT vs Other Tests

- LRT is asymptotically optimal
- For Normal, LRT = t-test or F-test
- Use $\nu = \#$ free params (unrestricted) - $\#$ free params (restricted)

Key Exam Problems

— Fall 2022 Final Exam —

P1 (2022): $Y_1, \dots, Y_n$ iid from $f(y) = \frac{1}{\theta} y^{\frac{1}{\theta}-1}$, $0 < y < 1$, $\theta > 0$. Find MLE of θ .

Likelihood: $L(\theta) = \prod_{i=1}^n \frac{1}{\theta} y_i^{\frac{1}{\theta}-1} = \theta^{-n} \prod_{i=1}^n y_i^{\frac{1}{\theta}-1}$

Log-likelihood: $\ell(\theta) = -n \log \theta + \left(\frac{1}{\theta} - 1 \right) \sum_{i=1}^n \log y_i$

Derivative: $\frac{d\ell}{d\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum \log y_i = 0$

Solve: $n\theta = -\sum \log y_i$

$$\hat{\theta}_{\text{MLE}} = -\frac{1}{n} \sum_{i=1}^n \log Y_i = -\overline{\log Y}$$

P2 (2022): $Y_1, \dots, Y_{25}$ iid $N(\mu, 9)$. Test $H_0 : \mu = 18$ vs $H_1 : \mu > 18$.

Rejection region: $\bar{Y} > 19$. Find power when $\mu = 20$.

Power = $P(\text{Reject } H_0 | \mu = 20)$

Under $\mu = 20$: $\bar{Y} \sim N(20, 9/25)$, $\text{SE} = \sqrt{9/25} = 0.6$

Standardize: $P(\bar{Y} > 19) = P\left(\frac{\bar{Y}-20}{0.6} > \frac{19-20}{0.6}\right)$
 $= P(Z > -1.667) = 1 - \Phi(-1.667) = \Phi(1.667)$

Using Z-table: Power = 0.9525

Note: Find Type I error by evaluating at $\mu = 18$

P3 (2022): $X_1, \dots, X_m$ iid $N(\mu_X, 1)$, $Y_1, \dots, Y_n$ iid $N(\mu_Y, 1)$

independent. Test $H_0 : \mu_X - \mu_Y = 2$ vs $H_1 : \mu_X - \mu_Y \neq 2$. Find rejection region $\bar{X} - \bar{Y} \notin [K_1, K_2]$ with Type I error α .

Construct test statistic under H_0 :

$\bar{X} - \bar{Y} \sim N(\mu_X - \mu_Y, \frac{1}{n} + \frac{1}{m})$

Under H_0 : $\mu_X - \mu_Y = 2$

Standardize: $Z = \frac{(\bar{X}-\bar{Y})-2}{\sqrt{1/m+1/n}} \sim N(0, 1)$

Two-sided test: Reject if $|Z| \geq z_{\alpha/2}$, i.e., $\bar{X} - \bar{Y} \notin [K_1, K_2]$

$$K_1 = 2 - z_{\alpha/2} \sqrt{\frac{1}{m} + \frac{1}{n}}$$

$$K_2 = 2 + z_{\alpha/2} \sqrt{\frac{1}{m} + \frac{1}{n}}$$

P4 (2022): $Y_1, \dots, Y_n$ iid $N(\mu, \sigma^2)$. Compare two estimators of μ : $\bar{Y}$ and $\hat{\mu}^* = \frac{1}{n+1} [\sum Y_i + 10]$.

(a) Find $\text{Var}(\bar{Y})$

$$\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n} \sum Y_i\right) = \frac{1}{n^2} \sum \text{Var}(Y_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

(b) Find $\text{Var}(\hat{\mu}^*)$

Rewrite: $\hat{\mu}^* = \frac{\sum Y_i + 10}{n+1} = \frac{n\bar{Y} + 10}{n+1}$

$\text{Var}(\hat{\mu}^*) = \text{Var}\left(\frac{n\bar{Y}}{n+1}\right) = \left(\frac{n}{n+1}\right)^2 \text{Var}(\bar{Y})$

$$= \frac{n^2}{(n+1)^2} \cdot \frac{\sigma^2}{n} = \frac{n\sigma^2}{(n+1)^2}$$

Note: $\frac{n\sigma^2}{(n+1)^2} < \frac{\sigma^2}{n}$ since $\frac{n}{(n+1)^2} < \frac{1}{n}$

(c) Find MSE for both estimators

For $\bar{Y}$: Unbiased, so $\text{MSE}(\bar{Y}) = \text{Var}(\bar{Y}) = \frac{\sigma^2}{n}$

For $\hat{\mu}^*$: $\mathbb{E}[\hat{\mu}^*] = \frac{n\mu+10}{n+1}$

Bias = $\frac{n\mu+10}{n+1} - \mu = \frac{n\mu+10-(n+1)\mu}{n+1} = \frac{10-\mu}{n+1}$

$\text{MSE}(\hat{\mu}^*) = \text{Var}(\hat{\mu}^*) + [\text{Bias}]^2 = \frac{n\sigma^2}{(n+1)^2} + \frac{(10-\mu)^2}{(n+1)^2}$

$$= \frac{n\sigma^2 + (10-\mu)^2}{(n+1)^2}$$

(d) Which is better if $\mu = 10$?

If $\mu = 10$: Bias = 0, so $\text{MSE}(\hat{\mu}^*) = \frac{n\sigma^2}{(n+1)^2}$

Compare: $\frac{n\sigma^2}{(n+1)^2}$ vs $\frac{\sigma^2}{n}$

$\frac{n\sigma^2}{(n+1)^2} < \frac{\sigma^2}{n} \iff \frac{n^2}{(n+1)^2} < 1$ (always true)

$\implies \hat{\mu}^*$ is better when $\mu = 10$

P5 (2022): $Y_1, \dots, Y_n$ iid $N(\mu, \sigma^2)$. Consider MLE $\hat{\sigma}^2 = \frac{1}{n} \sum (Y_i - \bar{Y})^2$.

(a) Find $\mathbb{E}[\hat{\sigma}^2]$ and $\text{Var}(\hat{\sigma}^2)$

Expectation: Recall $(n-1)S^2 = \sum (Y_i - \bar{Y})^2$ where $S^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$

Since $\mathbb{E}[S^2] = \sigma^2$:

$\mathbb{E}[\hat{\sigma}^2] = \mathbb{E}\left[\frac{1}{n} \sum (Y_i - \bar{Y})^2\right] = \frac{1}{n} \mathbb{E}[(n-1)S^2]$

$= \frac{n-1}{n} \mathbb{E}[S^2] = \frac{n-1}{n} \sigma^2$ (Biased! Underestimates by factor $\frac{n-1}{n}$)

Variance: Use fact that $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$\text{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \implies \text{Var}(S^2) = \frac{2(n-1)\sigma^4}{(n-1)^2} = \frac{2\sigma^4}{n-1}$

$\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{n-1}{n} S^2\right) = \left(\frac{n-1}{n}\right)^2 \text{Var}(S^2) = \frac{2(n-1)\sigma^4}{n^2}$

(b) Is $\hat{\sigma}^2$ consistent?

Check: Bias = $\frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n} \rightarrow 0$ as $n \rightarrow \infty$

$\text{Var}(\hat{\sigma}^2) = \frac{2(n-1)\sigma^4}{n^2} \sim \frac{2\sigma^4}{n} \rightarrow 0$ as $n \rightarrow \infty$

Both bias and variance $\rightarrow 0$, so YES, $\hat{\sigma}^2$ is consistent

P6 (2022): $Y \sim \text{Binomial}(n, p)$. MLE is $\hat{p} = Y/n$. Derive 95% CI for p using Fisher Information.

Step 1: Find Fisher Information

For Bernoulli(p): $I(p) = \frac{1}{p(1-p)}$ (single observation)

For n observations: $nI(p) = \frac{n}{p(1-p)}$

Step 2: MLE variance (Cramér-Rao)

$\text{Var}(\hat{p}) \approx \frac{1}{nI(p)} = \frac{p(1-p)}{n}$

Standard error: $\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

Step 3: Large sample CI

Asymptotic normality: $\hat{p} \xrightarrow{d} N\left(p, \frac{p(1-p)}{n}\right)$

Standardize: $\frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0, 1)$

Plug-in estimate: Replace p with $\hat{p}$ in SE

$$95\% \text{ CI: } \hat{p} \pm z_{0.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

— Fall 2023 Final Exam —

P1 (2023): Given $\mathbb{E}[X] = 3$, $\text{Var}(X) = 4$, $\mathbb{E}[Y] = 1$, $\text{Var}(Y) = 2$, $\text{Cov}(X, Y) = -1$. Let $W = 2Y - 4X + 1$. Find $\mathbb{E}[W]$ and $\text{Var}(W)$.

$$\mathbb{E}[W] = 2\mathbb{E}[Y] - 4\mathbb{E}[X] + 1 = 2(1) - 4(3) + 1 = \boxed{-9}$$

$$\text{Var}(W) = (-4)^2 \text{Var}(X) + (2)^2 \text{Var}(Y) + 2(-4)(2)\text{Cov}(X, Y) = 16(4) + 4(2) - 16(-1) = 64 + 8 + 16 = \boxed{88}$$

P2 (2023): Y has PDF $f(y) = \frac{1}{y\sqrt{2\pi}} e^{-(\log y - 2)^2/2}$, $y > 0$. Find distribution of $U = \log Y$.

Transformation: $U = \log Y \implies Y = e^U$, Jacobian: $\frac{dy}{du} = e^u$

$$f_U(u) = f_Y(e^u) \cdot e^u = \frac{1}{e^u \sqrt{2\pi}} e^{-(u-2)^2/2} \cdot e^u = \frac{1}{\sqrt{2\pi}} e^{-(u-2)^2/2}$$

$$U \sim N(2, 1)$$

P3 (2023): Population: $\mu = 3.4$, $\sigma = 0.5$ (known). Find minimum n for 95% CI width ≤ 0.2 .

$$\text{Width} = 2z_{0.025} \frac{\sigma}{\sqrt{n}} = \frac{1.96}{\sqrt{n}} \leq 0.2 \implies \sqrt{n} \geq 9.8 \implies n \geq 96.04, \text{ so}$$

$$\boxed{n = 97}$$

P4 (2023): $Y_1, \dots, Y_n$ iid $U(\theta, \theta + 1)$. Evaluate estimators.

(a) $\hat{\theta}_1 = \bar{Y}$: $\mathbb{E}[\bar{Y}] = \theta + 0.5$, Bias = 0.5 (NOT consistent)

(b) $\hat{\theta}_2 = Y_{(1)}$: $\mathbb{E}[Y_{(1)}] = \theta + \frac{1}{n+1}$, Bias $\rightarrow 0$ (consistent)

P5 (2023): $Y_1, \dots, Y_n$ iid $N(0, \sigma^2)$ ($\mu = 0$ known). Construct 95% CI for σ^2 .

$$\text{Pivotal: } Q = \frac{\sum Y_i^2}{\sigma^2} \sim \chi_n^2$$

$$P\left(\chi_{0.025, n}^2 \leq \frac{\sum Y_i^2}{\sigma^2} \leq \chi_{0.975, n}^2\right) = 0.95$$

$$95\% \text{ CI: } \left[\frac{\sum Y_i^2}{\chi_{0.975, n}^2}, \frac{\sum Y_i^2}{\chi_{0.025, n}^2} \right]$$

P6 (2023): $Y_1, \dots, Y_n$ iid from $f(y) = (2\beta + 2)y^{2\beta+1}$, $0 < y < 1$. Find MLE of β .

$$\ell(\beta) = n \log(2\beta + 2) + (2\beta + 1) \sum \log y_i$$

$$\frac{d\ell}{d\beta} = \frac{2n}{2\beta+2} + 2 \sum \log y_i = 0$$

$$\hat{\beta}_{\text{MLE}} = \frac{-n}{2 \sum \log y_i} - 1$$

— Fall 2024 Final Exam —

P1 (2024): $X_i \sim N(\mu, \sigma^2)$, $Y_i \sim N(\mu, 2\sigma^2)$ indep. Given $\hat{\sigma}^2 = \frac{1}{2n} \sum (X_i - \mu)^2 + \frac{1}{4n} \sum (Y_i - \mu)^2$.

$$\mathbb{E}[\hat{\sigma}^2] = \frac{1}{2n} \cdot n\sigma^2 + \frac{1}{4n} \cdot n(2\sigma^2) = \sigma^2 \text{ (Unbiased)}$$

$$\text{Var}(\hat{\sigma}^2) = \frac{\sigma^4}{2n} + \frac{\sigma^4}{2n} = \frac{\sigma^4}{n} \rightarrow 0 \text{ (Consistent)}$$

P2 (2024): $Y_1, \dots, Y_n$ iid from $f(y) = \frac{1}{\theta} y^{1/\theta}$, $0 < y < 1$, $\theta > 0$. Find MLE of θ .

$$\ell(\theta) = -n \log \theta + \frac{1}{\theta} \sum \log y_i$$

$$\frac{d\ell}{d\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum \log y_i = 0$$

$$\hat{\theta}_{\text{MLE}} = -\frac{1}{n} \sum \log Y_i$$

P3 (2024): $X_1, \dots, X_m$ iid $N(\theta, 1)$, $Y_1, \dots, Y_n$ iid $N(\theta + 1, 1)$ indep.

Construct 95% CI for θ using $\bar{X} + \bar{Y}$.

$$\bar{X} + \bar{Y} \sim N\left(2\theta + 1, \frac{1}{m} + \frac{1}{n}\right), \text{ so } \theta = \frac{\bar{X} + \bar{Y} - 1}{2}$$

$$95\% \text{ CI: } \frac{\bar{X} + \bar{Y} - 1}{2} \pm \frac{0.98}{\sqrt{1/m + 1/n}}$$

P4 (2024): Y has single observation from $U(3, 3\theta)$. Test $H_0: \theta = 4$ vs $H_1: \theta < 4$. Rejection region: $Y < c$ with $\alpha = 0.05$. Find c and power at $\theta = 3$.

Part (a): Find critical value c

Under $H_0: \theta = 4$: $Y \sim U(3, 12)$

$$\text{CDF under } H_0: F(y) = \frac{y-3}{12-3} = \frac{y-3}{9} \text{ for } 3 \leq y \leq 12$$

$$\text{Type I error: } \alpha = P(Y < c | H_0) = \frac{c-3}{9} = 0.05$$

$$\text{Solve: } c - 3 = 0.45 \implies \boxed{c = 3.45}$$

Part (b): Calculate power at $\theta = 3$

Under $H_1: \theta = 3$: $Y \sim U(3, 9)$

$$\text{CDF under } H_1: F(y) = \frac{y-3}{9-3} = \frac{y-3}{6} \text{ for } 3 \leq y \leq 9$$

$$\text{Power} = P(Y < 3.45 | \theta = 3) = \frac{3.45-3}{6} = \frac{0.45}{6} = \boxed{0.075}$$

P5 (2024): Two independent populations: $X \sim N(\mu_1, 9)$ and

$Y \sim N(\mu_2, 9)$. Samples: n from pop 1, $2n$ from pop 2. Find minimum n for 90% CI of $\mu_1 - \mu_2$ to have width ≤ 3 .

Step 1: Find variance of estimator

$\bar{X} - \bar{Y}$ is the estimator for $\mu_1 - \mu_2$

$$\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{2n}$$

$$= \frac{9}{n} + \frac{9}{2n} = \frac{18}{2n} + \frac{9}{2n} = \frac{27}{2n}$$

Step 2: CI width formula

$$90\% \text{ CI: } (\bar{X} - \bar{Y}) \pm z_{0.05} \sqrt{\frac{27}{2n}} \text{ where } z_{0.05} = 1.645$$

$$\text{Width} = 2z_{0.05} \sqrt{\frac{27}{2n}} = 2(1.645) \sqrt{\frac{27}{2n}} = 3.29 \sqrt{\frac{27}{2n}}$$

Step 3: Solve inequality

$$\text{Require: } 3.29 \sqrt{\frac{27}{2n}} \leq 3$$

$$\sqrt{\frac{27}{2n}} \leq \frac{3}{3.29} = 0.912$$

$$\frac{27}{2n} \leq (0.912)^2 = 0.8317$$

$$2n \geq \frac{27}{0.8317} = 32.46$$

$$n \geq 16.23$$

Therefore: $\boxed{n = 17}$ (round up to nearest integer)

Distribution	PMF/Range	Mean & Variance	Notes
Bernoulli(p)	$P(X = 1) = p, X \in \{0, 1\}$	$\mathbb{E}[X] = p, \text{Var}(X) = p(1-p)$	MGF: $(1-p) + pe^t$
Binomial(n, p)	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	$\mathbb{E}[X] = np, \text{Var}(X) = np(1-p)$	$X \in \{0, \dots, n\}$
Geometric(p)	$P(X = k) = (1-p)^{k-1} p$	$\mathbb{E}[X] = \frac{1}{p}, \text{Var}(X) = \frac{1-p}{p^2}$	Memoryless
Poisson(λ)	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$	$\mathbb{E}[X] = \lambda, \text{Var}(X) = \lambda$	$X \in \{0, 1, 2, \dots\}$
Distribution	PDF/Support	Mean & Variance	Notes
Uniform(a, b)	$f(x) = \frac{1}{b-a}, x \in [a, b]$	$\mathbb{E}[X] = \frac{a+b}{2}, \text{Var}(X) = \frac{(b-a)^2}{12}$	CDF: $\frac{x-a}{b-a}$
Normal(μ, σ^2)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\mathbb{E}[X] = \mu, \text{Var}(X) = \sigma^2$	$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$
Exponential(λ)	$f(x) = \lambda e^{-\lambda x}, x \geq 0$	$\mathbb{E}[X] = \frac{1}{\lambda}, \text{Var}(X) = \frac{1}{\lambda^2}$	Memoryless
Gamma(α, β)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$	$\mathbb{E}[X] = \frac{\alpha}{\beta}, \text{Var}(X) = \frac{\alpha}{\beta^2}$	$\chi_n^2 = \Gamma(n/2, 1/2)$
Beta(α, β)	$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$ $x \in [0, 1]$	$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}$ $\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$x \in [0, 1]$ Beta(1, 1) = $U(0, 1)$
Chi-squared(ν)	$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$	$\mathbb{E}[X] = \nu, \text{Var}(X) = 2\nu$	$\sum Z_i^2 \sim \chi_\nu^2$
t-distribution(ν)	$f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	$\mathbb{E}[X] = 0, \text{Var}(X) = \frac{\nu}{\nu-2}$	$\frac{Z}{\sqrt{V/\nu}} \sim t_\nu$
F(ν_1, ν_2)	Complex PDF	$\mathbb{E}[X] = \frac{\nu_2}{\nu_2-2} (\nu_2 > 2)$	$\frac{U/\nu_1}{V/\nu_2} \sim F_{\nu_1, \nu_2}$